

Relation Algebra

§16. Greedy Algorithms

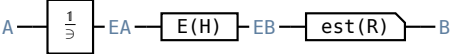
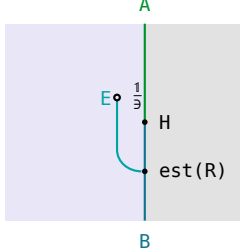
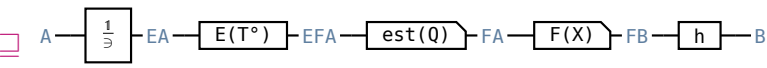
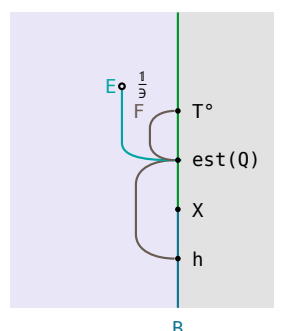
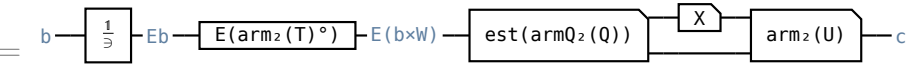
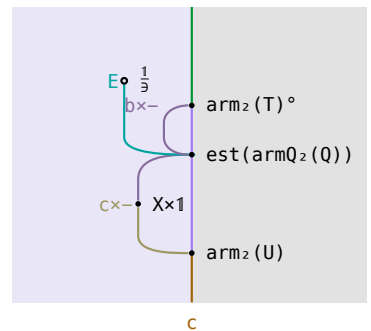
§16.1. Theory

(16.1a)

h, T, R, H, M as in $\triangle H$; additionally Q a **connected** preorder on the sets $\frac{T^\circ}{\mathfrak{S}}$ returns, so that $\frac{T^\circ}{\mathfrak{S}} \text{ est}(Q)$ is entire.

$$\text{PreservesRecip}(F) \wedge \text{Map}(h) \wedge \text{MonotonicAlg}(h, R) \wedge R R \subseteq R \wedge Q F(H) h \subseteq F(H) h R \implies (\mu X : \frac{T^\circ}{\mathfrak{S}} \text{ est}(Q) F(X) h) \subseteq \frac{H}{\mathfrak{S}} \text{ est}(R) \quad (16.1b)$$

the same optimum reached by keeping ONE decomposition at each step, so that no set is ever carried and the recursion runs on values alone

circuit	Hinze–Marsden
 the problem to be solved, $H \triangle (T)^\circ (h) \triangle H, Q$	
	
 Proposition 10.1 at $T = [V_1, V_2]$, $h = [U_1, U_2]$, $Q = Q_1 + Q_2$, $V_2 V_1^\circ = \emptyset$	

§16.2. The detab-entab problem

(16.2a)

$\text{detab} \triangle ([\text{nil}, \text{expand}]) : \text{String} \rightarrow \text{String}$ over snoc -lists, $\alpha \triangle [\text{nil}, \text{snoc}]$, $H = \text{detab}^\circ$; $\text{expand } (xs, a) = (a = \text{TB} \rightarrow \text{fill } xs, xs \# [a])$, $\text{fill } xs = xs \# \text{blanks } (n - (\text{col } xs) \bmod n)$.

$\text{col} \triangle ([\text{zero}, \text{count}])$, $\text{count } (c, a) = (a = \text{NL} \rightarrow 0, c + 1)$; TB the tab, BL the blank, NL the newline, tab stops every n columns.

$R \triangle \text{length} \leq \text{length}^\circ$, U the preorder with $a \cup b \iff a = \text{TB} v a = b$, $V \triangle \text{prefix}^\circ n (\text{fill } \text{fill}^\circ)$, $Q \triangle 1 + (V \times U)$.

$\text{unfill } xs$ the shortest prefix of xs with $\text{fill } (\text{unfill } xs) = \text{fill } xs$; tbc the trailing blank count, $\text{triple} \triangle (\text{unfill } \text{entab}, \{\text{tbc}, \text{col}\})$.

$$0 < n \wedge \text{blank} \neq n \Rightarrow (\mu X : \frac{[\text{nil}, \text{expand}]^\circ}{3} \text{est}(Q)F(X)[\text{nil}, \text{snoc}]) \sqsubseteq \frac{\text{detab}^\circ}{3} \text{est}(R), \text{ entab} = \text{triple} \quad (16.2b)$$

$$\text{assocl } \pi_1 (1 \times \text{blanks}) \text{ cat}$$

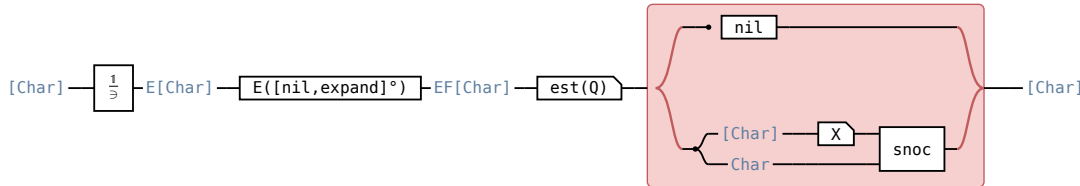
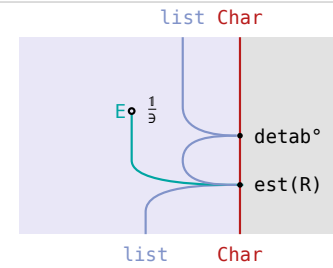
the shortest input **detab** expands to the given output is one pass along that output, holding each blank back and cashing the held blanks in for a tab wherever the column reaches a tab stop

circuit



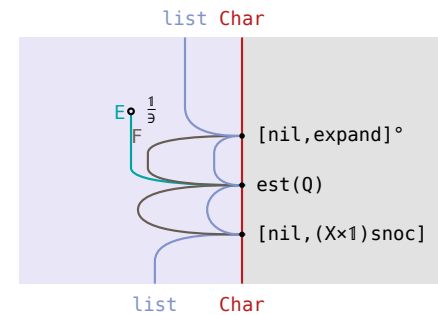
the specification $\text{---} \triangleq \text{detab}; \text{detab} \text{ entab}=1$ and nothing shorter does

Hinze–Marsden



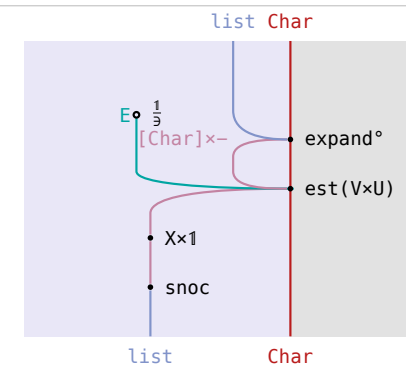
$\sqsubseteq$

at $Q \triangleq 1 + (V \times U)$: one character of input is decided at each step. $F(T, R) \alpha \sqsubseteq \alpha R$. **detab** $\text{prefix} \sqsubseteq R^\circ$ **detab** is FALSE, — at $n=8$, **detab** $[a, b, c, d, e, TB] = [a, b, c, d, e, BL, BL, BL]$, whose prefix $[a, b, c, d, e, BL, BL]$ is longer than any input giving it, and **detab** $V^\circ \sqsubseteq R^\circ$ **detab** holds. **expand** $V^\circ \sqsubseteq \text{expand} \cup (\pi_1 V^\circ)$ — shortening the output either leaves the last step alone or discards it



$=$

Proposition 10.1: **nil** and **expand** have disjoint ranges. The greedy step is to emit a tab whenever a tab is legal, consuming all the blanks back to the previous tab stop



$$\text{entab } xs = \text{entab } (\text{unfill } xs) \# \text{blanks } (\text{tbc } xs) \quad (10.1)$$

$=$

what makes $\text{triple} \triangleq (\text{unfill } \text{entab}, \text{tbc}, \text{col})$ a snoc-list reduce: the output splits at the last tab stop

$\text{triple} = ([\text{base}, \text{op}])$, $\text{entab} = \text{triple} \text{ assocl } \pi_1 (1 \times \text{blanks}) \text{ cat}$
base returns $([], (0, 0))$, **op** $((xs, (t, c)), a) =$

$=$

$(a = BL \wedge (c+1) \bmod n \neq 0 \rightarrow (xs, (t+1, c+1)), a = BL \rightarrow (xs \# [TB], (0, c+1)),$
 $a = NL \rightarrow (xs \# \text{blanks } t \# [NL], (0, 0)), (xs \# \text{blanks } t \# [a], (0, c+1)))$

the program: one pass carrying the column and the count of pending blanks

§16.3. The minimum tardiness problem

(16.3a)

$FX = 1 + (X \times \text{Job})$, $\alpha \triangleq [\text{nil}, \text{snoc}]$ on schedules, $\beta \triangleq [\text{nil}, \text{snag}]$ on bags, **snag** putting a job into a bag; $\text{bagify} \triangleq ([\beta]) : [\text{Job}] \rightarrow \text{Bag Job}$, $H = \text{bagify}^\circ$.

ct, **dt**, **wt** : $\text{Job} \rightarrow \text{Real}$ the completion, due and weighting quantities of a job; $\text{penalty}(xs, j) = (\text{sum}(\text{list}(\text{ct}(xs)) + \text{ct}(j) - \text{dt}(j)) \times \text{wt}(j))$.

$\text{cost} \triangleq \frac{\text{prefix}}{3} P(\alpha^\circ [\text{zero}, \text{penalty}]) \text{est}(\geq)$, $\text{cost } [] = 0$, $\text{cost } (xs \# [j]) = \text{bmax}(\text{cost } xs, \text{penalty}(xs, j))$, $R \triangleq \text{cost} \leq \text{cost}^\circ$.

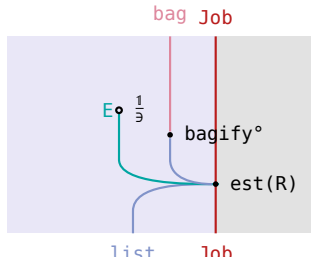
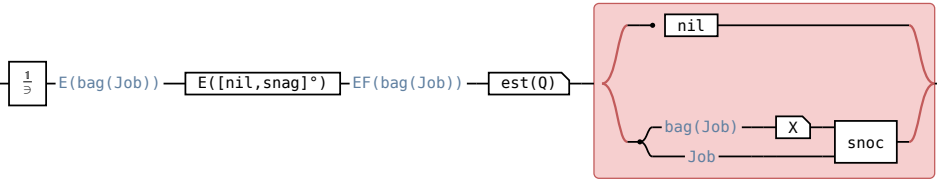
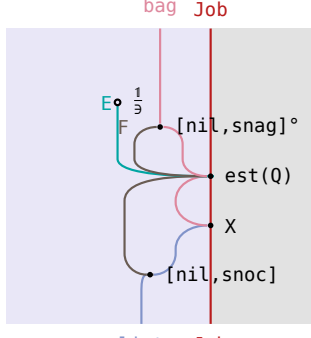
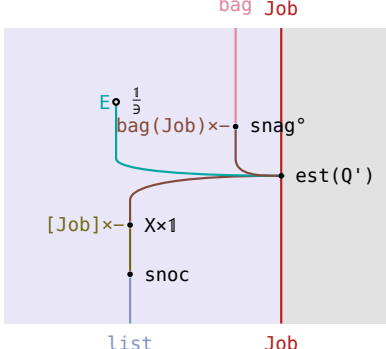
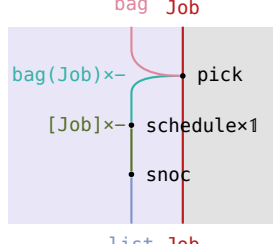
$\text{perm} \triangleq \text{bagify} \text{ bagify}^\circ = ([\text{nil}, \text{add}])$, $\text{add}(xs, j) = ys \# [j] \# zs$ for some $xs = ys \# zs$.

$k \triangleq [\text{zero}, \text{assocr } (1 \times ((\text{bagify}^\circ \times 1) \text{ penalty})) \text{ bmax}]$, $f \triangleq [\text{zero}, (\text{bagify}^\circ \times 1) \text{ penalty}]$, $Q \triangleq f \leq f^\circ$, $Q' \triangleq (\text{bagify}^\circ \times 1) \text{ penalty} \leq \text{penalty}^\circ (\text{bagify} \times 1)$.

$$(\forall j. 0 \leq \text{ct}(j)) \wedge (\forall j. 0 \leq \text{wt}(j)) \Rightarrow (\mu X : \frac{[\text{nil}, \text{snag}]^\circ}{3} \text{est}(Q) F(X) [\text{nil}, \text{snoc}]) \sqsubseteq_{\frac{\text{bagify}^\circ}{3}} \text{est}(R), \quad (16.3b)$$

schedule = (null → nil, pick (schedule × 1) snoc)

an ordering of the given bag with least maximum penalty is got by taking a job of least penalty out of the bag, putting it last, and scheduling what is left the same way

circuit	Hinze–Marsden
bag(Job) — $\frac{1}{3}$ — E(bag(Job)) — E(bagify[°]) — E[Job] — est(R) — [Job] the specification — $\triangleq$ bagify	
bag(Job) — $\frac{1}{3}$ — E(bag(Job)) — E([nil, snag][°]) — EF(bag(Job)) — est(Q) $\sqsubseteq$  No greedy reduce exists — one would also solve every prefix of the input, and the best schedule of a prefix need not extend to a best schedule of the whole	
$=$ bag(Job) — $\frac{1}{3}$ — E(bag(Job)) — E(snag[°]) — E(bag(Job) × Job) — est(Q') — X — snoc — [Job] Proposition 10.1: nil and snag have disjoint ranges	
$\sqsubseteq$ bag(Job) — pick — schedule — snoc — [Job] pick $\sqsubseteq_{\frac{\text{snag}^\circ}{3}} \text{est}(Q')$, a partial function, quadratic in the number of jobs	

§16.4. The TeX problem

(16.4a)

intern $\triangleq$ val round : Decimal $\rightarrow$ [0, 2¹⁶), val $\triangleq$ ([zero, shift]), shift (d, r) = (d + r) / 10, round r rounds 2¹⁶ r to the nearest integer: round r = n $\iff$ 2n - 1 < 2¹⁷ r < 2n + 1.

interval n = ((2n - 1) / 2¹⁷, (2n + 1) / 2¹⁷), r inrange (a, b) $\iff$ a < r < b, round[°] = interval inrange, R $\triangleq$ length $\leq$ length[°].

Interval all pairs (a, b), Legal(a, b) $\iff$ 0 < b < 1 and a < b (10.9), preserved by [arb, step] — arb_legal, step_legal [arb, step] : 1 + (Digit × Interval) $\rightarrow$ Interval, step (d, (a, b)) = ((d + a) / 10, (d + b) / 10).

FX = 1 + (Digit × X), $\alpha \triangleq$ [nil, cons], H $\triangleq$ ([arb, step])[°], ! : Digit × Interval $\rightarrow$ 1, Q $\triangleq$ (l[°] ![°] r) $\cup$ 1 l, r are coprod-laws's injections into FX = 1 + (Digit × X), so l : 1 $\rightarrow$ F(Interval) and r : Digit × Interval $\rightarrow$ F(Interval), w $\triangleq$ 2¹⁷.

$\text{interval}(\mu X : \frac{[\text{arb}, \text{step}]^\circ}{3} \text{ est}(Q) F(X) \alpha) \sqsubseteq \frac{\text{intern}^\circ}{3} \text{ est}(R) , \quad \text{extern}(n) = f(2n-1, 2n+1) \quad (16.4b)$ a shortest decimal whose internal representation is the given multiple of 2^{-16} is got by emitting the one digit the interval of admissible reals allows, until that interval contains zero and the empty decimal will do	
circuit	Hinze–Marsden
$[0, 2^{16}] \xrightarrow{\frac{1}{3}} E([0, 2^{16}]) \xrightarrow{E(\text{intern}^\circ)} E[\text{Digit}] \xrightarrow{\text{est}(R)} [\text{Digit}]$ the specification — $\triangleq$ intern	
$= [0, 2^{16}] \xrightarrow{\text{interval}} \text{Interval} \xrightarrow{\frac{1}{3}} E(\text{Interval}) \xrightarrow{E(\text{inrange val}^\circ)} E[\text{Digit}] \xrightarrow{\text{est}(R)} [\text{Digit}]$ round° is not a map, but interval is, so it comes out of the transpose	
$= [0, 2^{16}] \xrightarrow{\text{interval}} \text{Interval} \xrightarrow{\frac{1}{3}} E(\text{Interval}) \xrightarrow{E(H)} E[\text{Digit}] \xrightarrow{\text{est}(R)} [\text{Digit}]$ fusion: $\text{val inrange}^\circ = ([\text{arb}, \text{step}])$ — the converse of val, cut down to intervals, is a reduce on cons-lists	
$\sqsubseteq [0, 2^{16}] \xrightarrow{\text{interval}} \text{Interval} \xrightarrow{\frac{1}{3}} E(\text{Interval}) \xrightarrow{E([\text{arb}, \text{step}]^\circ)} EF(\text{Interval}) \xrightarrow{\text{est}(Q)} F(H) \xrightarrow{\alpha} [\text{Digit}]$ $\sqsubseteq$ $\frac{[\text{arb}, \text{step}]^\circ}{3}$ returns at most two elements — stop, or take one more digit — and $! \text{ nil} \sqsubseteq \text{cons } R^\circ$ makes it stop whenever stopping is legal	
$= \text{extern} = \text{interval } f, \quad f(a, b) = (a < 0 \rightarrow [], [d] \# f(10a-d, 10b-d))$ the program, with d the digit above	
$= \text{extern}(n) = f(2n-1, 2n+1), \quad f(p, q) = (p \leq 0 \rightarrow [], [d] \# f(10p-w \cdot d, 10q-w \cdot d))$ $d = (10q) \text{ div } w$: the same in integer arithmetic only, as chapter 3 required of intern — every interval reached is $(p/w, q/w)$	

§17. Appendix

§17.1. $P(S) \text{ est}(R) = (\exists S) n(\epsilon \setminus (SR^\circ))$

(17.1a)

(p, q) tabulates $W \triangleq (\exists S) n(\epsilon \setminus (SR^\circ))$, and $y \triangleq \frac{(p \exists S) n(qR)}{\exists}$, a map.

(17.1b)

$$\begin{array}{c} \boxed{\begin{array}{c} w \\ P(S) \text{ est}(R) \end{array}} \xleftarrow{p^\circ q = w, 1 \sqsubseteq yy^\circ} \begin{array}{c} \boxed{\begin{array}{c} p^\circ y \\ P(S) \end{array}} \\ \boxed{\begin{array}{c} y^\circ q \\ \text{est}(R) \end{array}} \end{array}$$

(17.1c)

$$\begin{array}{c} \boxed{\begin{array}{c} y^\circ q \\ \exists \end{array}} \xleftarrow{f^\circ \cdot \neg f^\circ, \cdot \exists \neg \exists} \boxed{\begin{array}{c} q \\ (p \exists S) n(qR) \end{array}} \xleftarrow{\Delta \neg n, f^\circ \cdot \neg f^\circ} \begin{array}{c} \boxed{\begin{array}{c} p^\circ q \\ \exists S \end{array}} \\ \boxed{\begin{array}{c} 1 \\ R \end{array}} \end{array}$$

R reflexive

$$\boxed{\begin{array}{c} \epsilon y^\circ q \\ R^\circ \end{array}} \xleftarrow{\cdot \exists \neg \exists, \circ, \text{meet}} \boxed{\begin{array}{c} R^\circ q^\circ q \\ R^\circ \end{array}} \quad \text{q simple}$$

(17.1d)

$$\begin{array}{c} \boxed{\begin{array}{c} p^\circ y \\ P(S) \end{array}} \xleftarrow{\Delta \neg n, \cdot T \neg / T, \circ, f^\circ \cdot \neg f^\circ} \begin{array}{c} \boxed{\begin{array}{c} p^\circ y \exists \\ \exists S \end{array}} \\ \boxed{\begin{array}{c} p \exists \\ y \exists S^\circ \end{array}} \end{array}$$

$$\boxed{\begin{array}{c} p^\circ y \exists \\ \exists S \end{array}} \xleftarrow{\cdot \exists \neg \exists, \text{meet}} \boxed{\begin{array}{c} p^\circ p \exists S \\ \exists S \end{array}} \quad \text{p simple}$$

$$\boxed{\begin{array}{c} p \exists \\ y \exists S^\circ \end{array}} \xleftarrow{\text{modular law}} \boxed{\begin{array}{c} p \exists \\ (p \exists) n(qRS^\circ) \end{array}} \xleftarrow{1 \sqsubseteq qq^\circ} \boxed{\begin{array}{c} q^\circ p \exists \\ RS^\circ \end{array}} \xleftarrow{\circ, T \cdot \neg T \setminus} \boxed{\begin{array}{c} w \\ \epsilon \setminus (SR^\circ) \end{array}} \quad \text{W's right half}$$

§17.2. $P(\text{est}(R)) \text{ est}(R) = P(\text{dom}(\text{est}(R))) \text{ union est}(R)$

(17.2a)

$$\begin{array}{c} \boxed{P(\text{est}(R)) \text{ est}(R)} \xrightarrow{\text{dom}(\text{est}(R)) \text{ est}(R) = \text{est}(R)} \boxed{P(\text{dom}(\text{est}(R)) \text{ est}(R)) \text{ est}(R)} \\ \xrightarrow{P \text{ a relator}} \boxed{P(\text{dom}(\text{est}(R))) P(\text{est}(R)) \text{ est}(R)} \xrightarrow{\text{est-711}} \boxed{P(\text{dom}(\text{est}(R))) \text{ union est}(R)} \end{array}$$

(17.2b)

$$\begin{array}{c} \boxed{\begin{array}{c} P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ P(\text{est}(R)) \text{ est}(R) \end{array}} \xleftarrow{(17.1b) \text{ at } S := \text{est}(R), \Delta \neg n, T \cdot \neg T \setminus} \boxed{\begin{array}{c} P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ \exists \text{est}(R) \end{array}} \\ \boxed{\begin{array}{c} \epsilon P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ \text{est}(R) R^\circ \end{array}} \xleftarrow{\text{count of } T \cdot \neg T \setminus} \boxed{\begin{array}{c} \exists (\exists n(\epsilon \setminus (\epsilon \setminus (SR^\circ)))) \\ \exists (\exists n(\epsilon \setminus (SR^\circ))) \end{array}} \\ \xleftarrow{\text{modular law}} \boxed{\begin{array}{c} (\exists \exists) n(\epsilon \setminus (\epsilon \setminus (SR^\circ))) \\ \exists \text{est}(R) \end{array}} \xleftarrow{T(UnV) \sqsubseteq TUnTV, \cdot \exists \neg \exists, \text{est-73}} \boxed{\begin{array}{c} P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ \exists \text{est}(R) \end{array}} \\ \xleftarrow{P(\text{dom}(\text{est}(R))) \sqsubseteq 1} \boxed{\begin{array}{c} \text{union est}(R) \\ \exists \text{est}(R) \end{array}} \\ \xleftarrow{\epsilon \text{ lax natural}} \boxed{\begin{array}{c} \epsilon P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ \text{est}(R) R^\circ \end{array}} \xleftarrow{\text{dom}(\text{est}(R)) \epsilon \text{ union est}(R)} \boxed{\begin{array}{c} \text{dom}(\text{est}(R)) \epsilon \text{ union est}(R) \\ \text{est}(R) R^\circ \end{array}} \\ \xleftarrow{\text{est}(R) \epsilon \text{est}(R)} \boxed{\begin{array}{c} \text{est}(R) \text{ est}(R)^\circ \epsilon \text{ union est}(R) \\ \text{est}(R) R^\circ \end{array}} \xleftarrow{\text{dom}(\text{est}(R)) \sqsubseteq \text{est}(R) \text{ est}(R)^\circ} \boxed{\begin{array}{c} \text{est}(R) \epsilon \text{est}(R) \\ \text{est}(R) R^\circ \end{array}} \quad \text{UP of est}$$

§17.3. Shortest paths on a cylinder, on lists

(17.3a)

definition	type	note
$F(A, X) = A + A \times X$	$\mathcal{A} \rightarrow \mathcal{A}$	The base functor: one square of the new column, alone or in front of the path so far.
$L = \text{list}^+$	$\mathcal{A} \rightarrow \mathcal{A}$	A path is a non-empty list of squares, one per column crossed.
α the initial algebra	$F(\text{dCL}(L, E)) \rightarrow \text{dCL}(L, E)$	$\alpha(5) = [5]$ and $\alpha(1, [5]) = [1, 5]$: a path is started by one square, or extended by one.
N the n-tuple relator	$\mathcal{A} \rightarrow \mathcal{A}$	One component per row of a column, so the fold carries n answers at once.
$R \triangleq \text{sum} \leq \text{sum}^\circ$	$[\text{Nat}] \rightarrow [\text{Nat}]$	The cost of a path, which the cheapest minimises.
setify	$A[n] \rightarrow EA$	$\text{setify}(1, 2, 3, 4) = \{1, 2, 3, 4\}$ — which row a component came from is forgotten.
moves	$A[n] \rightarrow E(A[n])$	$\text{moves}(x) = \{\text{up}(x), x, \text{down}(x)\}$ — rotated up, unrotated, rotated down.
trans	$E(A[n]) \rightarrow (E A)[n]$	$\text{trans}\{(a, b, c), (x, y, z)\} = (\{a, x\}, \{b, y\}, \{c, z\})$ — component k of the result is the set of the k-th components.
zip	$F(a[n]) \rightarrow (F a)[n]$	$\text{zip}((1, 2, 3, 4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})) = ((1, \{[5]\}), (2, \{[6]\}), (3, \{[7]\}), (4, \{[8]\}))$.
$\text{cp} \triangleq \frac{F(1, \exists)}{\exists}$	$F(EA) \rightarrow EFA$	$\text{cp}(1, \{[5], [6], [8]\}) = \{(1, [5]), (1, [6]), (1, [8])\}$.
$\text{gen} \triangleq F(1, \text{moves trans } N(\text{union})) \text{zip } N(\text{cp}(F(A, -), T)E(\alpha))$	$F(NA, N(E(TA))) \rightarrow N(E(TA))$	$\text{gen}((1, 2, 3, 4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\}))$ is worked out in (17.3.1b).
$\text{paths} \triangleq (\text{gen}) \text{setify union}$	$T(NA) \rightarrow E(TA)$	$\text{paths}[(1, 2, 3, 4), (5, 6, 7, 8)]$ is the union of (17.3.1b)'s four sets: 12 paths, 3 from each entry row.
the specification $\text{paths est}(R)$	$[\text{Nat}[n]] \rightarrow [\text{Nat}]$	A cheapest path from the entry side to the exit side.

§17.3.1. $N(E(L A))$

$$\begin{array}{l} 1 \quad 5 \\ 2 \quad 6 \\ 3 \quad 7 \\ 4 \quad 8 \end{array} \quad [(1, 2, 3, 4), (5, 6, 7, 8)] : L \ N \ \text{Nat}$$

(17.3.1a)

$$(\text{gen})[(5, 6, 7, 8)] = (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})$$

(17.3.1b)

$$\begin{aligned} \text{gen}((1, 2, 3, 4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})) = \\ (\{ [1, 5], [1, 6], [1, 8] \}, \\ \{ [2, 5], [2, 6], [2, 7] \}, \\ \{ [3, 6], [3, 7], [3, 8] \}, \\ \{ [4, 5], [4, 7], [4, 8] \}) \end{aligned}$$

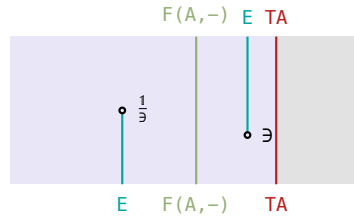
(17.3.1c)

N	the 4-tuple, one component per row of the entry column
E	the set of paths that can start in that row
L	a path is the non-empty list of the squares it crosses

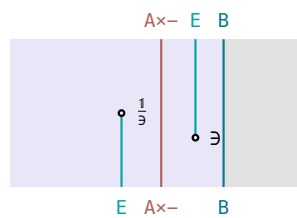
row 1's set has no $[1,7]$ — from row 1 only rows 4, 1 and 2 are reachable — and it does have $[1,8]$, because the cylinder glues the bottom row to the top. That is **moves trans N(union)** of $\triangleq$ **paths**: component k collects the paths of rows $k-1, k, k+1$.

§17.3.2. The cross product $cp \triangleq \frac{F(1, \exists)}{\exists}$

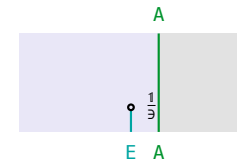
(17.3.2a)



cp on all of $F(A, E(L A))$



the $A \times -$ summand: $\exists$ picks one path, $1\% \exists$ collects the results



the A summand, no E to distribute: $1\% \exists$ alone, $a \mapsto \{a\}$

(17.3.2b)

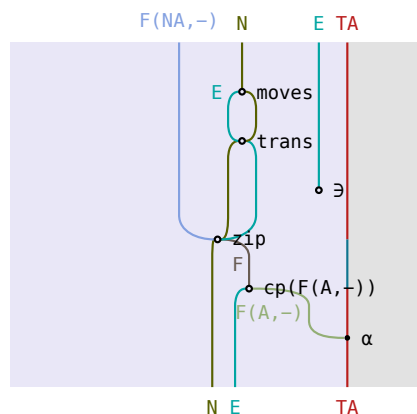
type	note
$F(A, -) = A + A \times -$	one wire carries the whole functor, both summands with it; the $\exists$ under it is $F(1, \exists)$
$\exists : E(L A) \rightarrow L A$	the $\exists$ inside $F(1, \exists)$: a set of paths, one of them — $L A$ is one path, A one square
$F(1, \exists) = 1 + 1 \times \exists$	$: A + A \times E(L A) \rightarrow A + A \times L A$, the relator acting on each summand
$\exists : E(A + A \times L A) \rightarrow A + A \times L A$	the $\exists$ under the bar: a set of cells, one of them — a different $\exists$
$cp : A + A \times E(L A) \rightarrow E(A + A \times L A)$	$\frac{R}{\exists}$ turns $R : X \rightarrow Y$ into $X \rightarrow E Y$
$cp = \frac{1}{\exists} E(F(1, \exists))$	what the picture draws: the unit makes the outer E , leaving one $\exists$ — pow-laws

(17.3.2c)

$w = (1, \{[5], [6], [8]\}) : A \times E(L A)$ the right summand
 $1 \times \exists : A \times E(L A) \rightarrow A \times L A$ 1 keeps the square 1, $\exists$ picks one path
 $w \mapsto (1, [5]), (1, [6]), (1, [8]) : A \times L A$ one output per path in w
 $cp(w) = \{(1, [5]), (1, [6]), (1, [8])\} : E(A + A \times L A)$ the three of them, collected

§17.3.3. $gen = F(1, \text{moves trans } N(\text{union})) \text{ zip } N(cp P(\alpha))$

(17.3.3a)

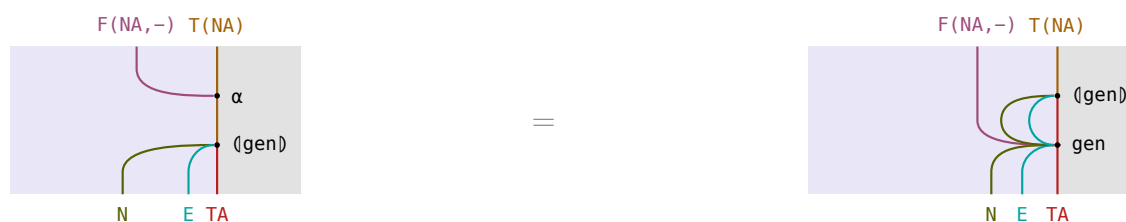


(17.3.3b)

$u = ((1,2,3,4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\}))$: $F(N A, N(E(L A)))$
 $F(1, \text{moves trans } N(\text{union}))$: 1 keeps the column, the path SETS move
 $\text{moves}(\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})$
 $= \{(\{[6]\}, \{[7]\}, \{[8]\}, \{[5]\}),$
 $(\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\}),$
 $(\{[8]\}, \{[5]\}, \{[6]\}, \{[7]\})\}$: $E(N(E(L A)))$ down, unmoved, up
 $\text{trans}(\text{that})$
 $= (\{[6]\}, \{[5]\}, \{[8]\}, \{[7]\}, \{[6]\}, \{[5]\}),$
 $\{[8]\}, \{[7]\}, \{[6]\}, \{[5]\}, \{[8]\}, \{[7]\})$: $N(E(E(L A)))$ row k gets rows k-1, k, k+1
 $N(\text{union})(\text{that})$
 $= (\{[5], [6], [8]\}, \{[5], [6], [7]\},$
 $\{[6], [7], [8]\}, \{[5], [7], [8]\})$: $N(E(L A))$ every path into row k, none dropped
 $\text{zip}(\text{that})$
 $= ((1, \{[5], [6], [8]\}), (2, \{[5], [6], [7]\}),$
 $(3, \{[6], [7], [8]\}), (4, \{[5], [7], [8]\}))$: $N(F(A, E(L A)))$ each row: its square, and the paths it may be put in front of
 $N(\text{cp } P(\alpha))(\text{that})$
 $= (\{[1,5], [1,6], [1,8]\}, \{[2,5], [2,6], [2,7]\},$
 $\{[3,6], [3,7], [3,8]\}, \{[4,5], [4,7], [4,8]\})$: $N(E(L A))$ cp pairs the square with each path, α prefixes it: $\alpha(1, [5]) = [1,5]$

§17.3.4. gen is an F-algebra; $(\text{gen}) : \alpha(\text{gen}) = F(1, (\text{gen}))\text{gen}$

(17.3.4a)



α puts the column back on the list, then the fold reads all of it

the fold reads the rest under F , then one gen puts the column in front

$$xs = [(1,2,3,4), (5,6,7,8)] = \alpha((1,2,3,4), [(5,6,7,8)]) : L(N A)$$

(17.3.4b)

$$(\text{gen})(xs) = \text{gen}((1,2,3,4), (\text{gen})[(5,6,7,8)])$$

both halves are in (17.3.1b): the tail folds to $(\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})$, one path per row and each of them one square long, and gen on it is (17.3.3b)'s walk.

§17.3.5. $Q : F(N A, N(L A)) \rightarrow N(L A)$, an $F(N A, -)$ -algebra: $Q = F(1, \text{moves trans } N(\text{est}(R))) \text{ zip } N(\alpha)$

(17.3.5a)

$Q : F(N A, N(L A)) \rightarrow N(L A)$: the fold's algebra: a new column, one path per row
 $u = ((1,2,3,4), ([5], [6], [7], [8]))$: $F(N A, N(L A))$
 $F(1, \text{moves trans } N(\text{est}(R)))$: 1 keeps the column, the paths move
 $\text{moves}([5], [6], [7], [8])$
 $= (\{[6], [7], [8], [5]\},$
 $\{[5], [6], [7], [8]\},$
 $\{[8], [5], [6], [7]\})$: $E(N(L A))$ down, unmoved, up
 $\text{trans}(\text{that})$
 $= (\{[6], [5], [8]\}, \{[7], [6], [5]\},$
 $\{[8], [7], [6]\}, \{[5], [8], [7]\})$: $N(E(L A))$ row k gets rows k-1, k, k+1
 $N(\text{est}(R))(\text{that})$
 $= ([5], [5], [6], [5])$: $N(L A)$ the cheapest into each row – chosen BEFORE the new square
 $\text{zip}(\text{that})$
 $= ((1, [5]), (2, [5]), (3, [6]), (4, [5]))$: $N(F(A, L A))$ each row: its square, and the one predecessor that survived
 $N(\alpha)(\text{that})$
 $= ([1,5], [2,5], [3,6], [4,5])$: $N(L A)$ $\alpha(1, [5]) = [1,5]$

$Q = [N(\text{wrap}), (1 \times \text{moves trans } N(\text{est}(R))) \text{ zip}' N(\text{cons})]$ (17.3.5b)	
at the last column Q starts one path per row, and at each earlier one it puts each square in front of the cheapest of the three kept paths it can step to — the algebra (17.3.5c)'s last step folds	
circuit — the fork is $F(NA, N(LA)) = NA + NA \times N(LA)$	Hinze–Marsden
$F(N A) (N (T A)) \text{---} \boxed{Q} \text{---} N(T A)$	
$FN(T A) \text{---} \boxed{F(1, \text{moves trans } N(\text{est}(R)))} \text{---} FN(T A) \text{---} \boxed{\text{zip}} \text{---} N(F A (T A)) \text{---} \boxed{N(\alpha)} \text{---} N(T A)$ the fusion condition (17.3.5d) read as a definition,	
$F(N A) (N (T A)) \text{---} \left\{ \begin{array}{l} N(p_{\circ}) \text{---} \boxed{N(\text{wrapA})} \\ q \text{---} \boxed{\text{pm}} \text{---} q \text{---} \boxed{\text{zip}'} \text{---} N(q_{\circ}) \text{---} \boxed{N(\text{consA})} \end{array} \right\} \text{---} N(T A)$ $\text{zip} = 1 + \text{zip}', \alpha = [\text{wrap}, \text{cons}]$	

paths $\text{est}(R) \exists (Q)$ setify $\text{est}(R)$ (17.3.5c)	
shortest paths on a cylinder: the dynamic programming approach to exhaustive search allows a path of least cost to be found in $O(n \times m)$ time; Q is (17.3.5b)'s algebra.	
circuit	Hinze–Marsden
$T(N\ A) \text{---} \boxed{\text{paths}} \text{---} E(T\ A) \text{---} \boxed{\text{est}(R)} \text{---} T\ A$	
$= T(N\ A) \text{---} \boxed{F(N\ A)\ (N\ (E\ (T\ A)))} \text{---} \boxed{\text{gen}} \text{---} NE(T\ A) \text{---} \boxed{\text{setify}} \text{---} EE(T\ A) \text{---} \boxed{\text{union}} \text{---} E(T\ A) \text{---} \boxed{\text{est}(R)} \text{---} T\ A$ $\triangleq \text{paths at paths}$	
$\sqsubseteq T(N\ A) \text{---} \boxed{F(N\ A)\ (N\ (E\ (T\ A)))} \text{---} \boxed{\text{gen}} \text{---} NE(T\ A) \text{---} \boxed{\text{setify}} \text{---} EE(T\ A) \text{---} \boxed{P(\text{est}(R))} \text{---} E(T\ A) \text{---} \boxed{\text{est}(R)} \text{---} T\ A$ $P(\text{est}(R)) \text{ est}(R) \sqsubseteq \text{union est}(R) \text{--- a minimum in each set, then a minimum of those; } R \text{ transitive}$	
$\sqsubseteq T(N\ A) \text{---} \boxed{F(N\ A)\ (N\ (E\ (T\ A)))} \text{---} \boxed{\text{gen}} \text{---} NE(T\ A) \text{---} \boxed{N(\text{est}(R))} \text{---} N(T\ A) \text{---} \boxed{\text{setify}} \text{---} E(T\ A) \text{---} \boxed{\text{est}(R)} \text{---} T\ A$ $\text{setify lax natural}$	
$\sqsubseteq T(N\ A) \text{---} \boxed{F(N\ A)\ (N\ (T\ A))} \text{---} \boxed{Q} \text{---} N(T\ A) \text{---} \boxed{\text{setify}} \text{---} E(T\ A) \text{---} \boxed{\text{est}(R)} \text{---} T\ A$ $\text{fusion at (17.3.5d)}$	

gen $N(\text{est}(R)) \exists F(1, N(\text{est}(R))) Q$ (17.3.5d)	
fusion: the condition for fusion in (17.3.5c)'s last step, used to derive a definition of Q .	
circuit	Hinze–Marsden
$F(N\ A)\ (N\ (E\ (T\ A))) \text{---} \boxed{\text{gen}} \text{---} NE(T\ A) \text{---} \boxed{N(\text{est}(R))} \text{---} N(T\ A)$	
$\sqsubseteq FNE(T\ A) \text{---} \boxed{F(1, N(\text{est}(R)))} \text{---} FN(T\ A) \text{---} \boxed{Q} \text{---} N(T\ A)$ $(7.13), \text{ then zip, trans, moves lax natural}$	

(7.13) is $F(1, \text{est}(R)) \alpha \sqsubseteq_{cp} P(\alpha) \text{ est}(R)$, mon-thm71 at the map α with $\frac{F(1, \exists) \alpha}{\exists} =_{cp} P(\alpha)$: extending every path in a set and then taking a minimum is beaten by extending one minimum. It is the crux here, not the greedy theorem.